DARK MODE PREFERENCE AND ITS IMPACT ON USER COMFORT

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ABSTRACT

The increasing use of digital devices in education, work, and daily communication has led to prolonged screen exposure, raising concerns related to eye strain, visual fatigue, and overall user comfort. To address these concerns, interface design features such as dark mode have been widely introduced across operating systems and applications. Dark mode employs light-colored text on dark backgrounds and is believed to reduce screen glare and visual discomfort, particularly in low-light environments. However, user perceptions regarding its effectiveness remain varied, and empirical evidence—especially in the local context—remains limited.

This study aimed to determine the perceived impact of dark mode on user comfort, specifically in terms of readability, eye strain, and overall visual comfort. A quantitative descriptive research design was employed, and data were gathered through an online survey questionnaire distributed via Google Forms. Respondents consisted of digital device users with prior experience using dark mode on smartphones, laptops, and desktop computers. The questionnaire utilized a five-point Likert scale to measure user perceptions, and data were analyzed using descriptive statistical tools such as frequency, percentage distribution, and weighted mean.

The findings revealed that most respondents preferred dark mode, particularly during nighttime use and in low-light environments. Results further indicated that dark mode was perceived to be effective in reducing eye strain and minimizing screen glare, contributing to a more comfortable and relaxed viewing experience. However, readability yielded comparatively lower weighted mean scores, suggesting that user comfort in dark mode varies depending on individual preferences, device type, and interface design factors. Overall, the study concludes that dark mode has a positive perceived effect on user comfort, although its effectiveness is influenced by contextual and individual differences. The results of this study may serve as a basis for future research and provide insights for developers and users in optimizing display preferences for improved visual ergonomics.

I. INTRODUCTION

The rapid advancement of digital technology has significantly transformed the way people communicate, work, study, and access information. With the widespread use of smartphones, laptops, and desktop computers, individuals are increasingly exposed to digital screens for extended periods. This prolonged screen exposure has raised concerns regarding eye strain, visual fatigue, headaches, and other discomforts associated with excessive screen use. As digital devices become integral to daily life, ensuring user comfort and visual well-being has become an important consideration in interface and system design.

To mitigate the adverse effects of prolonged screen exposure, various display and interface design solutions have been introduced, one of which is dark mode. Dark mode is a display setting that uses light-colored text, icons, and graphical elements on a dark background. It has gained widespread popularity due to its perceived benefits, such as reduced screen brightness, minimized glare, and enhanced visual comfort, particularly in low-light environments. Many operating systems, applications, and websites now offer dark mode as an alternative to the traditional light mode, allowing users to customize their viewing experience based on their preferences.

Despite its growing adoption, the effectiveness of dark mode in improving user comfort remains a subject of debate. Some studies suggest that dark mode can reduce eye strain and visual fatigue, while others argue that light mode may offer better readability and visual clarity, especially for prolonged reading tasks. These conflicting findings indicate that user comfort is influenced by multiple factors, including lighting conditions, screen brightness, font type, contrast ratio, and individual visual sensitivity. Consequently, user preference for dark mode may vary depending on context and personal needs.

In the Philippine setting, where online learning, remote work, and digital communication have become increasingly prevalent, there is limited local research that examines the impact of dark mode on user comfort. Understanding how users perceive dark mode in terms of readability and visual comfort is essential in promoting healthier screen usage and user-centered interface design. Anchored in Visual Ergonomics Theory and Human–Computer Interaction (HCI) Theory, this study seeks to determine the perceived effect of dark mode on user comfort among digital device users. Specifically, it aims to assess user preferences and perceptions related to eye strain, readability, and overall visual comfort.

The findings of this study are expected to contribute to existing literature by providing empirical evidence on the perceived benefits and limitations of dark mode. Moreover, the results may guide users in making informed decisions regarding display settings and assist developers, designers, and educators in creating more accessible, comfortable, and user-friendly digital environments.

II. EXPERIMENTAL METHOD/S

Research Design

The study employed a quantitative descriptive research design to determine the perceived effect of dark mode on user comfort. This design was deemed appropriate as it allows the systematic collection and analysis of numerical data to describe trends, preferences, and perceptions of respondents without manipulating any variables.

IPO Model

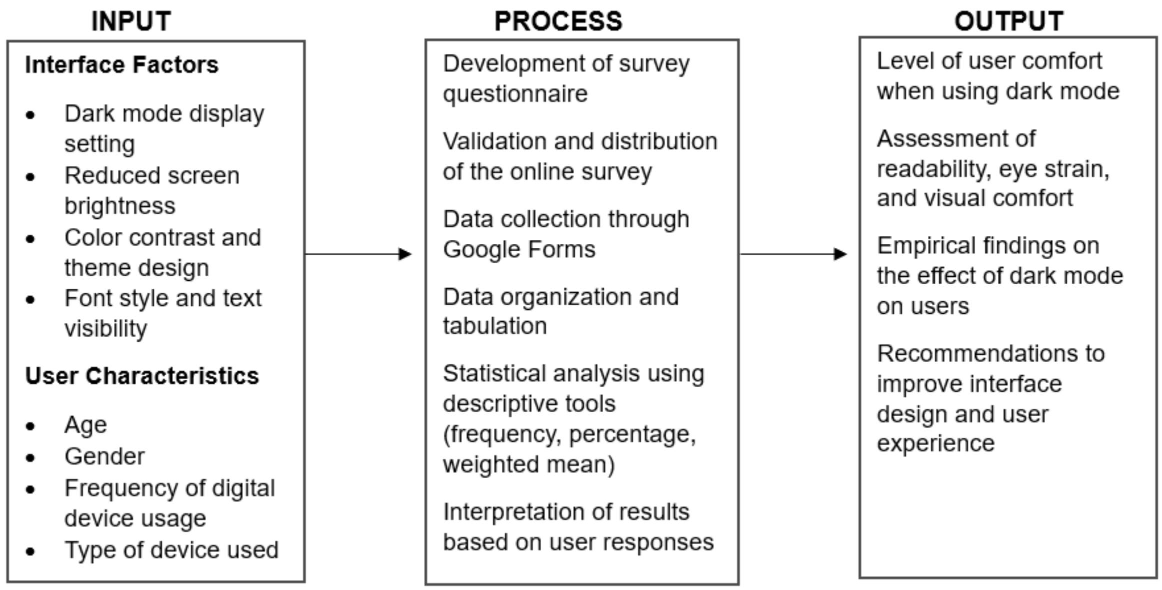

Figure 1 illustrates the Input–Process–Output (IPO) model of the study. The input consists of interface factors such as dark mode display setting, screen brightness, color contrast, and font visibility, as well as user characteristics including age, gender, frequency of digital device usage, and type of device used. The process includes the development and validation of the survey questionnaire, data collection through Google Forms, data organization and tabulation, and statistical analysis using descriptive tools such as frequency, percentage, and weighted mean. The output represents the level of user comfort when using dark mode, including assessments of readability, eye strain, and visual comfort, as well as empirical findings and recommendations for improving interface design and user experience.

Sources of Data and Research Instruments

The primary source of data for this study was obtained from individuals who regularly use digital devices and have prior experience using dark mode as a display setting. These respondents served as the main source of information in determining perceptions related to readability, eye strain, and overall visual comfort. The respondents were selected using convenience sampling, as participation depended on accessibility and willingness to respond to the online survey. Data were gathered using a structured survey questionnaire administered through Google Forms. The questionnaire was composed of two parts: the first part collected demographic information such as age, gender, frequency of digital device usage, and type of device used, while the second part consisted of statements that measured user comfort when using dark mode in terms of readability, eye strain, and visual comfort. A five-point Likert scale ranging from Strongly Disagree to Strongly Agree was used to quantify respondents' perceptions and ensure consistency in data measurement.

Tools for Data Analysis

The collected data were systematically organized, encoded, and tabulated using spreadsheet software to ensure accuracy and clarity of analysis. Descriptive statistical tools were employed to analyze the responses gathered from the respondents. Frequency counts and percentage distribution were used to describe the demographic profile of the respondents and identify trends in digital device usage and dark mode preference. The weighted mean was utilized as the primary analytical tool to determine the overall level of user comfort when using dark mode, allowing for the interpretation of respondents' level of agreement for each indicator related to readability, eye strain, and visual comfort.

Statistical Treatment of Data

The statistical treatment of data involved the use of descriptive statistical techniques to objectively interpret the responses of the respondents. Frequency and percentage distribution were applied to summarize the demographic characteristics of the respondents, while the weighted mean was computed for each statement in the questionnaire to determine the general tendency of responses. Numerical weights were assigned to each Likert scale option, and the resulting mean scores were interpreted using a corresponding verbal interpretation scale. These statistical procedures enabled a systematic evaluation of user comfort when using dark mode and served as the basis for the discussion of findings and formulation of conclusions.

III. RESULTS AND DISCUSSION

Table 1 presents the summary of the survey results on user comfort when using dark mode based on the computed weighted mean and corresponding verbal interpretation. The criteria reflected in the table were aligned with the indicators used in the survey questionnaire to assess readability, eye strain, and overall visual comfort.

Table 1. Survey Results on User Comfort When Using Dark Mode

Criteria	**Mean**	**Verbal Interpretation**
Readability of text when using dark mode	4.32	Agree
Reduction of eye strain when using dark mode	4.45	Strongly Agree
Visual comfort during prolonged screen use	4.38	Agree
Reduction of screen glare	4.47	Strongly Agree
Comfort when using dark mode in low-light environments	4.50	Strongly Agree
Overall comfort when using dark mode	4.41	Agree
Overall Composite Mean	**4.42**	**Agree**

Readability of text when using dark mode evaluated how clearly and comfortably respondents were able to read on their digital devices, and it obtained a mean score of 4.32, which indicated an Agree level of perception. This result suggests that most users found text readable in dark mode, although readability may still vary depending on individual preferences and display settings. Reduction of eye strain, which measured the perceived effectiveness of dark mode in minimizing visual fatigue during screen use, earned a high mean score of 4.45, indicating a Strongly Agree level of agreement. This implies that dark mode is highly effective in alleviating eye strain among users. Visual comfort during prolonged screen use achieved a mean score of 4.38, interpreted as Agree, indicating that respondents generally experienced comfort when using dark mode for extended periods.

Reduction of screen glare, which assessed the ability of dark mode to lessen excessive brightness and glare, obtained a mean score of 4.47, interpreted as Strongly Agree, suggesting that dark mode significantly improves viewing conditions, especially in low-light environments. Comfort when using dark mode in low-light environments recorded the highest mean score of 4.50, also interpreted as Strongly Agree, demonstrating that respondents strongly perceived dark mode as suitable and comfortable under dim lighting conditions. Overall comfort when using dark mode garnered a mean score of 4.41, interpreted as Agree, indicating a positive general perception of dark mode as a display feature.

The composite mean of 4.42, verbally interpreted as Agree, reflects an overall positive assessment of user comfort when using dark mode. This result indicates that, when all indicators are considered collectively, respondents consistently expressed favorable perceptions regarding the comfort and visual benefits provided by dark mode. Overall, the survey results confirm that dark mode contributes positively to user comfort by reducing eye strain, minimizing glare, and enhancing visual comfort during digital device usage.

IV. CONCLUSION

The appraisal of the study exhibited a high level of user comfort across all evaluated indicators, where the mean scores ranged from 4.32 to 4.50, indicating a generally positive perception of dark mode in terms of readability, eye strain reduction, visual comfort, and usability in low-light environments. This finding indicates that dark mode effectively contributes to improved visual comfort and a more pleasant viewing experience during digital device usage.

Prolonged exposure to digital screens was found to significantly affect users in their daily activities, particularly in tasks that require extended reading, concentration, and continuous screen interaction. Visual discomfort, eye strain, and excessive screen glare may negatively influence productivity and overall well-being. Additionally, discomfort caused by inappropriate display settings can lead to fatigue and reduced engagement in academic, professional, and leisure activities.

Dark mode presents a practical solution to these challenges by providing a display option that reduces brightness and glare while enhancing visual comfort, especially in low-light environments. The ability to view content with reduced eye strain allows users to maintain focus, minimize visual fatigue, and improve overall screen-related experience. By offering an alternative display mode that accommodates varying visual needs, dark mode supports healthier and more comfortable digital interaction.

Overall, the findings suggest that dark mode has a positive impact on user comfort and should be maintained as a customizable feature in digital interfaces. Allowing users to select display settings that suit their visual preferences can promote better visual ergonomics, enhance user satisfaction, and support long-term digital device use in a wide range of contexts.

V. REFERENCES

Buchner, A., & Baumgartner, N. (2007). Text–background polarity affects performance irrespective of ambient illumination and colour contrast.

https://www.tandfonline.com/doi/abs/10.1080/00140130701306413

Nielsen Norman Group. (2020). Dark mode: Benefits and drawbacks.

https://www.nngroup.com/articles/dark-mode/

Nielsen Norman Group. (2023). Dark mode: How users think about it and issues to avoid.

https://www.nngroup.com/articles/dark-mode-users-issues

Pathari, F. J. (2024). Dark vs. Light Mode on Smartphones: Effects on Eye Fatigue. ACHI 2024 Conference Proceedings. https://www.thinkmind.org/articles/achi_2024_3_150_20069.pdf

Android Developers. (n.d.). Dark theme.

https://developer.android.com/develop/ui/views/theming/darktheme

Sengsoon, P., & Intaruk, R. (2025). Immediate effects of light mode and dark mode features on visual fatigue in tablet users. International Journal of Environmental Research and Public Health.

https://www.mdpi.com/1660-4601/22/4/609

Laine, J. (2025). Impact of Dark Mode on User Experience and Eye Comfort. Theseus Digital Repository. https://www.theseus.fi/bitstream/handle/10024/896088/Laine_Jere.pdf